



Surya Workshop Datasets and Dataloaders

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SOLAR SYSTEM SCIENCE & EXPLORATION

Thank you very much to AWS



Goals

- Teach the community how to build and train AI applications.
- Get all started in using Surya.
- Write a joint publication with both positive and negative results, as well as lessons learned.
- Provide structure so that development is not overwhelming.

Sharing resources

- We will all be sharing resources today and tomorrow. Expect some jank and discomfort.
- Please coordinate with your team on who is launching runs in your assigned GPU.
- Start small and focus on end-to-end. Let's plan on scaling up on the afternoon of Wednesday.
- Let's experiment with short 5 minute help slots, in combination with AI assistants.

Template overview

Data creation and surya indices

Official Surya release

```
✓ SURYA_WORKSHOP
> 📁 data
> 📁 downstream_apps \ template
> 📁 Surya
> 📁 workshop_infrastructure
🔴 .gitignore
🔴 .gitmodules
📄 environment.yml
🔑 LICENSE
📄 README.md
```

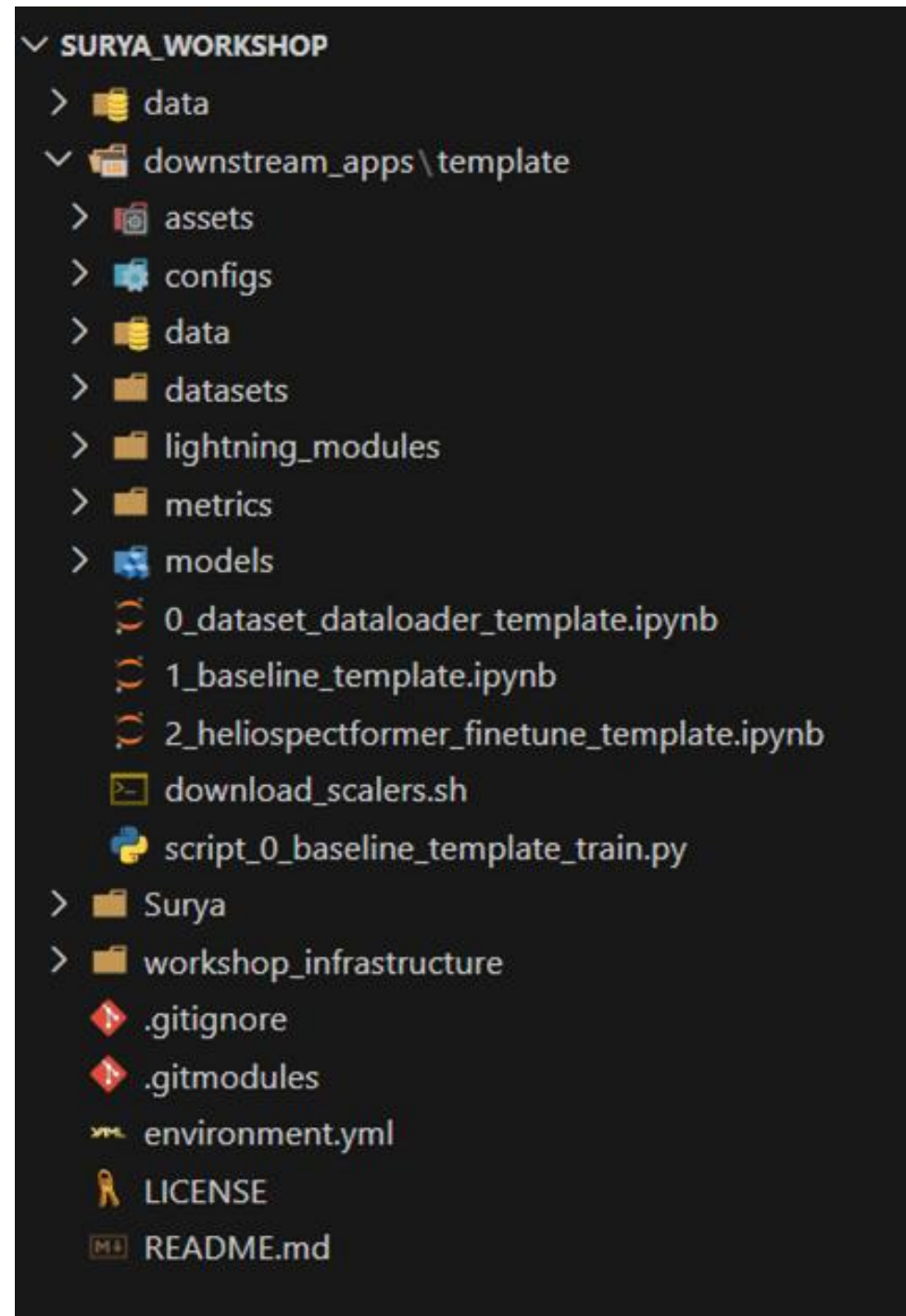
Downstream template

Modifications to release for AWS

Template overview

Data creation and surya indices

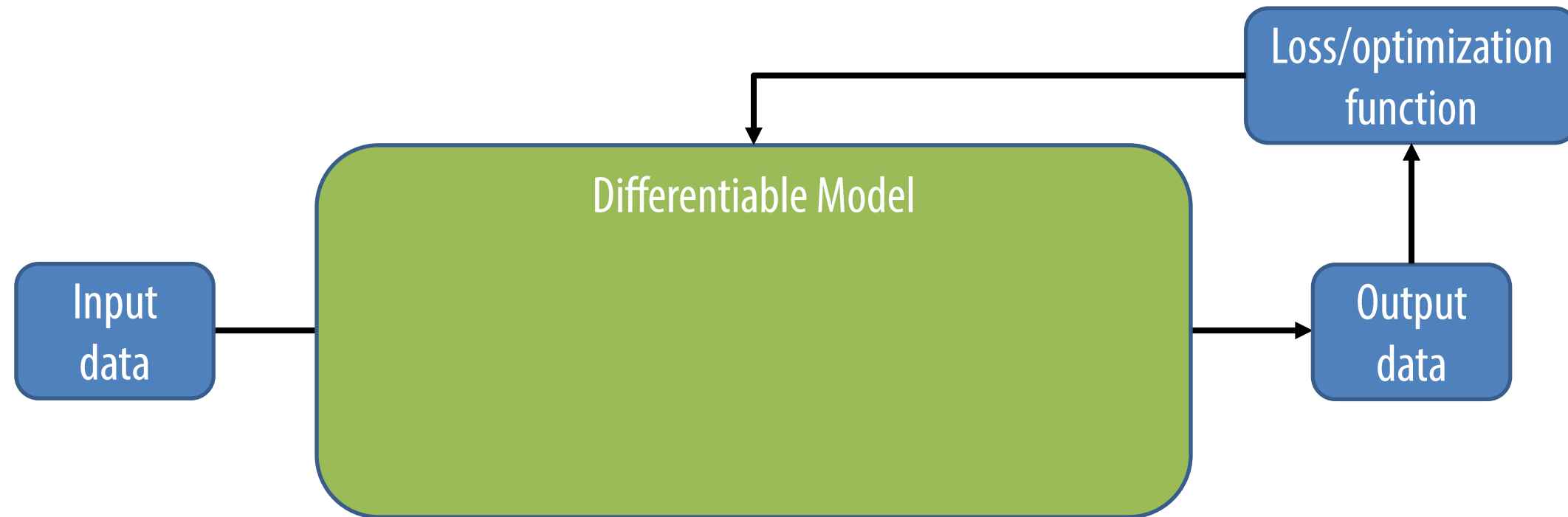
Official Surya release



Downstream template

Modifications to release for AWS

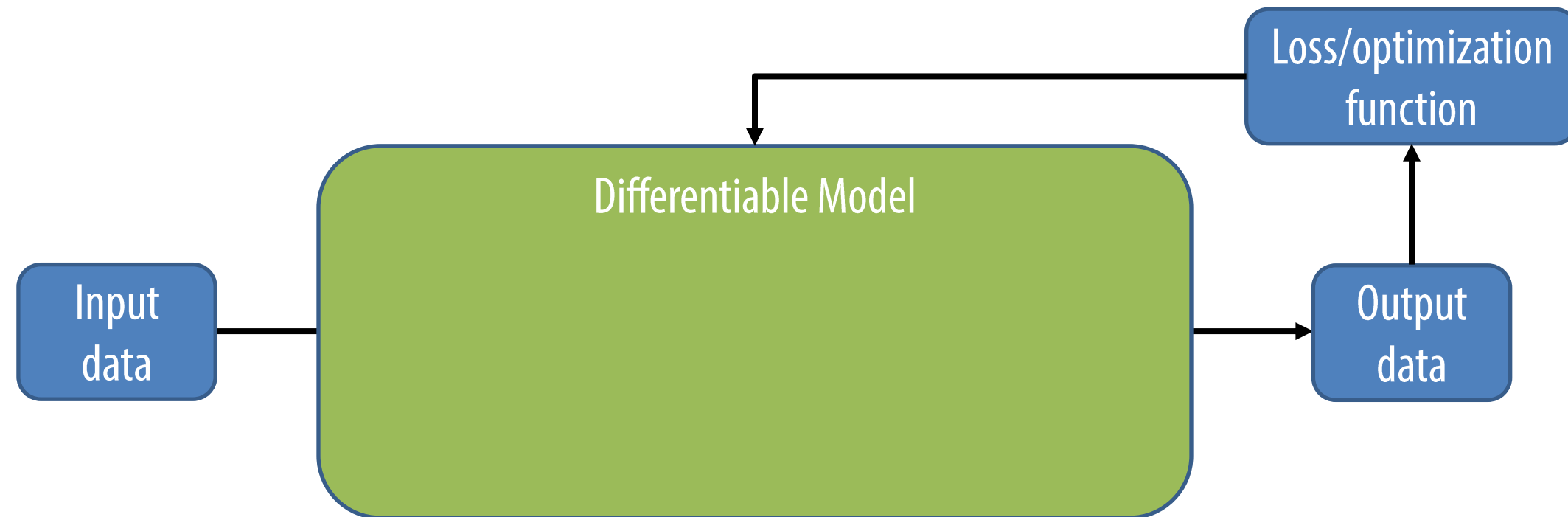
The elements of an AI training loop



Dataset

vs.

Dataloader



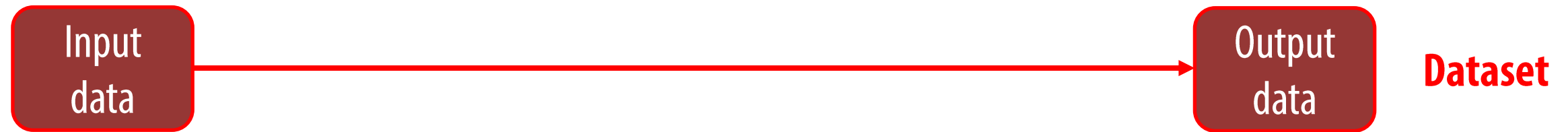
Dataset

vs.

Dataloader

- Indexes and structures data.
- Is deterministic (enumerates data).
- It's where all the work goes.

- Reads and manages the data.
- Defines a sampling strategy (can be random).
- It's a simple wrapper



Basic Elements of a Dataset

- `__init__`
 - Loads and processes indices.
 - Sets all necessary variables.
- `__len__`
 - Returns the number of data.
- `__getitem__`
 - Returns a single input-target pair.
 - Always the same pair for a given index.

```
1 class FaceLandmarksDataset(Dataset):
2     """Face Landmarks dataset."""
3
4     def __init__(self, csv_file, root_dir, transform=None):
5         """
6         Arguments:
7             csv_file (string): Path to the csv file with annotations.
8             root_dir (string): Directory with all the images.
9             transform (callable, optional): Optional transform to be applied
10                on a sample.
11         """
12         self.landmarks_frame = pd.read_csv(csv_file)
13         self.root_dir = root_dir
14         self.transform = transform
15
16     def __len__(self):
17         return len(self.landmarks_frame)
18
19     def __getitem__(self, idx):
20         if torch.is_tensor(idx):
21             idx = idx.tolist()
22
23         img_name = os.path.join(self.root_dir,
24                                self.landmarks_frame.iloc[idx, 0])
25         image = io.imread(img_name)
26         landmarks = self.landmarks_frame.iloc[idx, 1:]
27         landmarks = np.array([landmarks], dtype=float).reshape(-1, 2)
28         sample = {'image': image, 'landmarks': landmarks}
29
30         if self.transform:
31             sample = self.transform(sample)
32
33         return sample
```

Surya's Dataset

- Takes an index of data points in the form of a CSV.
- Defines the channels to use and return and the nature of the time sampling.
- Returns dictionaries with datapoint info.

See Surya's dataset:

https://github.com/SwRI-IDEA-Lab/surya_workshop/blob/main/workshop_infrastructure/datasets/helio.py



Template dataset

- Child class of Surya's dataset.
- Takes a downstream application index and finds intersection with available Surya data.
- Adds downstream information to Surya's dictionary.

See the template dataset:

https://github.com/SwRI-IDEA-Lab/surya_workshop/blob/main/downstream_apps/template/datasets/template_dataset.py

